

No. Variables

What is python Variables?

what is Global Variables?

what is global Keyword?

How to get Id of any object?

a=1, b=1 does both have same Id or not?

What is None in Python

Difference between None, NaN, and Null

How is memory managed in Python?

Ques. What is python Variables?

- Variables are containers for storing data values.
- A variable name must **start** with a **letter** or the **underscore** character.
- A variable name **cannot** start with a **number**.
- Variable names are **case-sensitive** (age, Age and AGE are three different variables)

```
#Legal variable names:
myvar = "John"
MYVAR = "John"
my_var = "John"
myVar = "John"
myvar2 = "John"
_my_var = "John"

#Illegal variable names:
2myvar = "John"
my-var = "John"
my var = "John"
```

```
x = 5
y = "Mohit"
print(x) # Output:- 5
print(y) # Output:- Mohit
```

Single or Double Quotes:-

- In Python, single quotes (' ') and double quotes (" ") are functionally the same and both are used to create strings. The choice mainly depends on readability and avoiding unnecessary escaping when the string contains quotes.

```
x = "John"
print(x) # Output:- John

# double quotes are the same as single quotes:
x = 'John'
print(x) # Output:- John

print('He said "Hello"') # Output:- He said "Hello"
print("It's a beautiful day")
print('It\'s a beautiful day')
print('It's a beautiful day') # Output:- error
```

Assign Multiple Values:

- Python allows you to assign values to multiple variables in one line.

```
x, y, z = "Orange", "Banana", "Cherry"

print(x) # Output:- Orange
print(y) # Output:- Banana
print(z) # Output:- Cherry
```

One Value to Multiple Variables:

- we can assign the same value to multiple variables in one line.

```
x = y = z = "Orange"

print(x) # Output:- Orange
print(y) # Output:- Orange
print(z) # Output:- Orange
```

- Variables Casting:** We want to specify the data type of a variable, this can be done with casting. and We can **get the data type** of a variable with the **type()** function.

```
x = str(3)
y = int(3)
z = float(3)

print(x)
print(y)
print(z)

Output:-
3
3
3.0
```

```
x = 5
y = "John"
print(type(x))
print(type(y))

Output:-
<class 'int'>
<class 'str'>
```

- **Unpack a Collection:** If we have a collection of values in a list, tuple etc. Python allows you to extract the values into variables. This is called unpacking.

```
fruits = ["apple", "banana", "cherry"]  
x, y, z = fruits
```

```
print(x)  
print(y)  
print(z)
```

Output:-

```
apple  
banana  
cherry
```

Output Variables(combine both text and a variable)

```
x = "awesome"  
print("Python is " + x)
```

output:- Python is awesome

Example 2

```
x = "Python is "  
y = "awesome"  
z = x + y  
print(z)
```

output:-Python is awesome

```
x = 5  
y = 10  
print(x + y)
```

output:- 15

Note:- If you try to combine a string and a number, Python will give you an error:

```
x = 5  
y = "John"  
print(x + y)  
output:- TypeError: unsupported operand type(s) for +: 'int' and 'str'
```

Ques. Global Variables?

- Variables that are created outside of a function.
- Global variables can be used by everyone, both inside of functions and outside.

```
x = "awesome"
def myfunc():
    print("Python is " + x)
myfunc()
```

output:- Python is awesome

```
x = "awesome"
def myfunc():
    x = "fantastic"
    print("Python is " + x)
myfunc()
print("Python is " + x)
```

output:-

Python is fantastic

Python is awesome

Ques. global Keyword?

- To create a global keyword inside a function should be treated as a global variable, you can use the **global keyword**.

```
def myfunc():  
    global x  
    x = "fantastic"  
  
myfunc()  
print("Python is " + x) # Output:- Python is fantastic
```

- Also, use the global keyword if you want to change a global variable inside a function.

```
x = 10 # Global variable  
def modify_global():  
    global x  
    x = 20 # Changing the value of the global variable  
  
modify_global()  
print(x) # This will print 20
```

Ques. How to get Id of any object?

- The id() function returns the unique identity of an object during its lifetime.
 - It returns an integer
 - This value is unique and constant for the object while it exists
 - **id()** function takes a single parameter object.

```
a = 5  
print(id(a))
```

Ques. a=1, b=1 does both have same Id or not?

- Two variables in Python have same id,
- but not in lists because they are stored in different memory locations.

```
a = 10  
b = 10  
print(id(a)) # Output:- 9788992  
print(id(b)) # Output:- 9788992
```

- In List

```
a = [1, 2, 3]
b = [1, 2, 3]
print(a is b) # Output:- Output:- False
```

- In tuples

```
a = (1, 2, 3)
b = (1, 2, 3)
print(a is b) # Output:- True
```

What is None in Python?

- None in Python represents the absence of a value and is commonly used to indicate that a variable has no assigned value or a function returns nothing.
- Important Rule:- Always use is / is not with None.

```
# Always compare with None using:
if x is None:

# Avoid:
if x == None:
```

Difference between None, NaN, and Null

1. None
 1. Represents **absence of a value**
 2. An **object** of type NoneType
2. NaN (Not a Number)
 1. Represents invalid or undefined numeric value
 2. Comes from math operations
 3. Used in floating-point calculations
 4. Common in NumPy / Pandas / Data Science
3. Null
 1. NOT used in Python, Used in Java, JavaScript, PHP, C, C++

Ques. How is memory managed in Python?

- Memory management in Python is handled by the **Python Memory Manager**.
- Python also has an inbuilt garbage collector, which recycles all the unused memory and so that it can be made available to the heap space.
- Memory management in python is managed by **Python private heap space**. All Python objects and data structures are located in a private heap. The programmer does not have access to this private heap. The python interpreter takes care of this instead.

- The allocation of heap space for Python objects is done by Python's memory manager. The core API gives access to some tools for the programmer to code.